Employee Attrition Analytics Platform - Final Technical Report

Note: the XGBoost row was produced with scikit-learn GradientBoosting (class-weighted) as a stand-in because the XGBoost library was unavailable where this document was generated. Running Part 1 in Google Colab regenerates this report with real XGBoost results (the note will not appear).

1. Business Problem and Objectives

Employee attrition is costly (recruitment, onboarding, lost knowledge). In the IBM HR Analytics dataset, 237 of 1470 employees (16.1%) left. The objective is an end-to-end HR analytics platform that (1) predicts who is at risk of leaving, (2) compares neural networks with classical, Bayesian and discriminant methods, (3) deploys the best model as a REST scoring service on IBM Watson Machine Learning with automated retraining, and (4) presents HR KPIs and risk insights in a dashboard.

2. HR Analytics Workflow and Architecture

- Data: IBM HR Analytics CSV (1,470 rows, 35 columns, no missing values).
- Preparation: remove constant/ID columns, scale numeric features, one-hot encode categoricals, stratified 80/20 split.
- Modelling: seven models from four families, compared with 5-fold cross-validation and a held-out test set.
- Selection: best model chosen by cross-validated ROC-AUC, refitted on all employees and saved.
- Scoring: out-of-fold risk scores converted into Low / Medium / High risk categories.
- Deployment: Watson Studio notebook stores the model in Watson Machine Learning and exposes an online REST endpoint.
- Automation: a Watson Studio Job retrains on a schedule, applies a quality gate (AUC) and updates the same endpoint.
- Dashboard: HR KPIs, department trends, risk categories, demographics and model metrics, fed by exported data files.

3. Data and Preprocessing

Features used: 30 (after removing EmployeeCount, Over18, StandardHours and EmployeeNumber). The classes are imbalanced, so accuracy alone is misleading; recall, F1 and ROC-AUC are emphasised. Imbalance is handled with class weights (Logistic Regression, Random Forest), scale_pos_weight (XGBoost), random oversampling of the training data (neural network) and equal priors (LDA/QDA).

4. Models Compared

- Classical ML: Logistic Regression, Random Forest, XGBoost.
- Neural network: multilayer perceptron with two hidden layers (64 and 32 units), L2 regularisation and early stopping.
- Bayesian classification: Gaussian Naive Bayes. It applies Bayes' theorem, P(Leave | features) = P(features | Leave) x P(Leave) / P(features), assuming independent features. The learned prior was 0.162; the model updates this prior into a posterior probability for each employee.
- Discriminant analysis: LDA (shared covariance) and QDA (class-specific covariance, regularised). They were applied to numeric features only, because they assume continuous, roughly normal inputs, which one-hot columns violate.

5. Results

Model	Family	Accuracy	Precision	Recall	F1	ROC-AUC	CV ROC-AUC
Logistic Regression	Classical ML	0.772	0.381	0.681	0.489	0.812	0.831
Random Forest	Classical ML	0.840	0.500	0.277	0.356	0.786	0.806
XGBoost	Classical ML	0.864	0.733	0.234	0.355	0.809	0.821
Neural Network (MLP)	Neural Network	0.847	0.521	0.532	0.526	0.730	0.812
Gaussian Naive Bayes	Bayesian	0.646	0.261	0.660	0.373	0.706	0.764
LDA	Discriminant	0.667	0.262	0.596	0.364	0.702	0.753
QDA	Discriminant	0.612	0.244	0.681	0.360	0.694	0.747

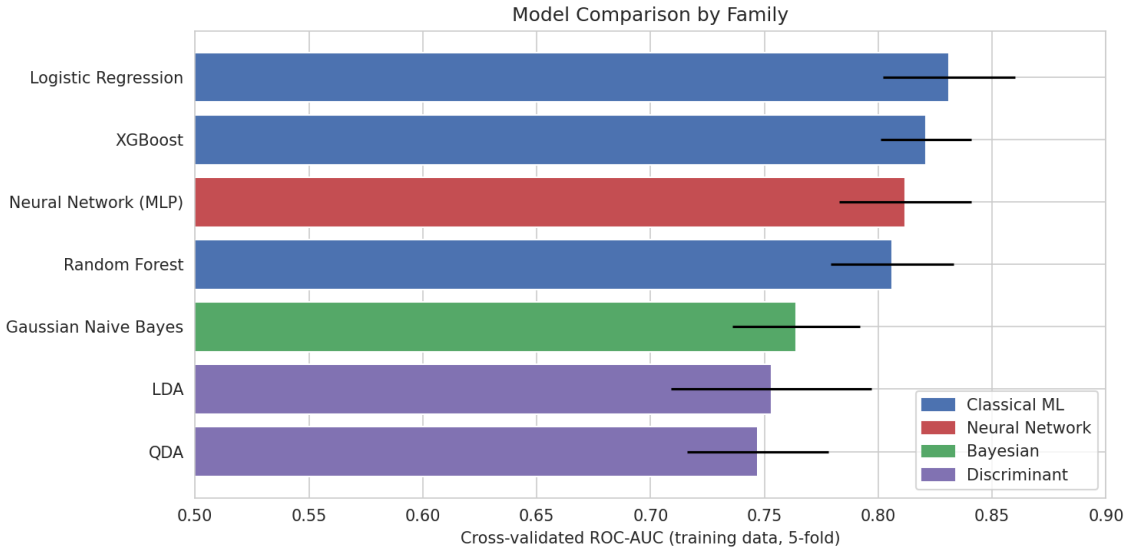


Figure 1: Cross-validated ROC-AUC by model family

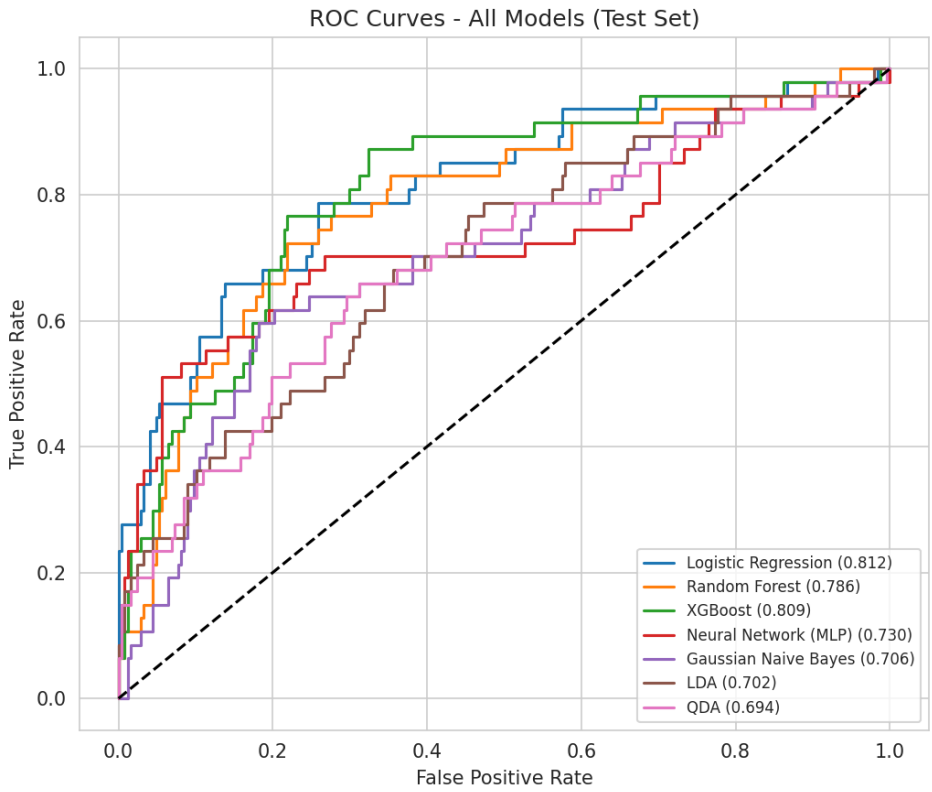


Figure 2: ROC curves of all models on the test set

5.1 Neural Network versus Classical ML

The neural network reached a cross-validated ROC-AUC of 0.812, compared with 0.831 for the best classical model (Logistic Regression). It did not outperform the classical model. With only 1,470 employees and mostly tabular, roughly linear relationships, a simple regularised model is hard to beat, and the neural network is also harder to explain to HR.

5.2 Bayesian and Discriminant Analysis

Gaussian Naive Bayes (CV AUC 0.764) and LDA/QDA (best CV AUC 0.753) are valuable for interpretability and as baselines, but their independence and normality assumptions are only partly met in HR data, so they rank below the best classical and neural models.

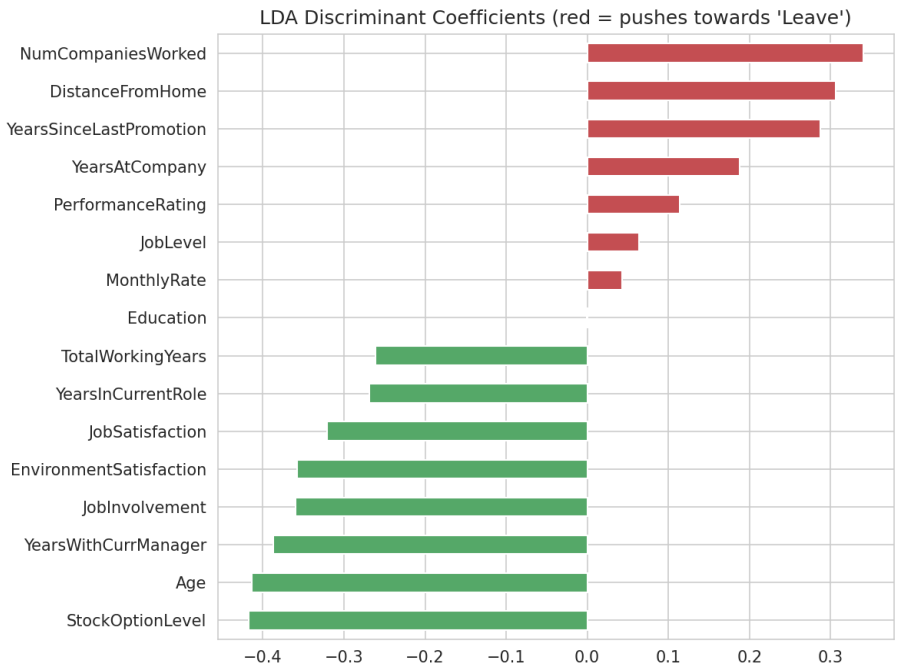


Figure 3: LDA discriminant coefficients

6. Selected Model

Logistic Regression (Classical ML) was selected because it had the highest cross-validated ROC-AUC (0.831). On the untouched test set: Accuracy 0.772, Precision 0.381, Recall 0.681, F1 0.489, ROC-AUC 0.812. Confusion matrix: 32 leavers caught, 15 missed, 52 false alarms, 195 stayers correctly identified.

Top drivers (permutation importance, drop in ROC-AUC when the feature is shuffled): OverTime (0.106), YearsSinceLastPromotion (0.027), NumCompaniesWorked (0.026), MaritalStatus (0.019), JobSatisfaction (0.019), BusinessTravel (0.018).

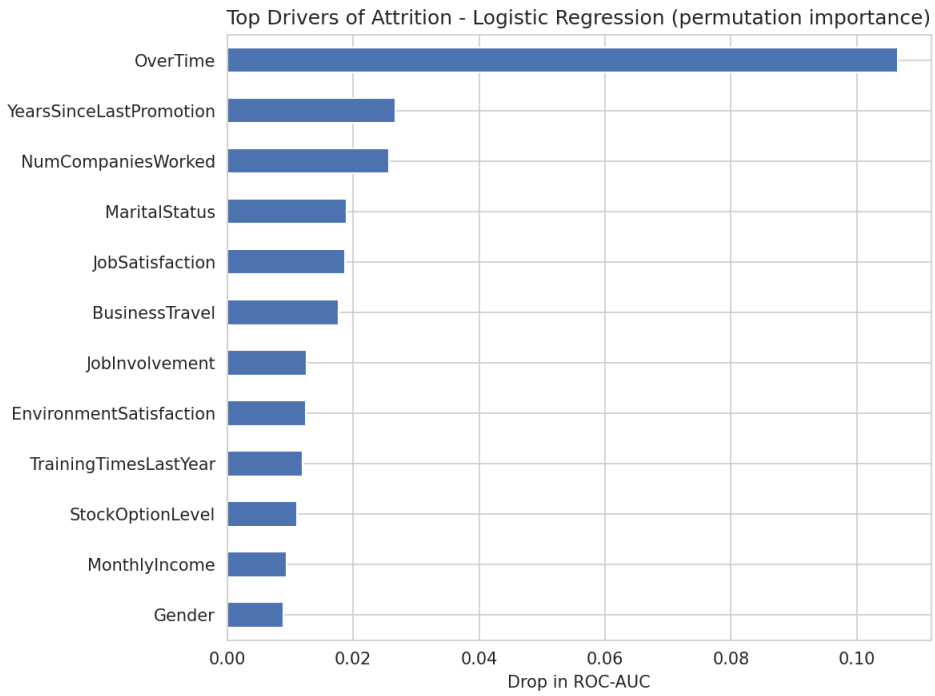


Figure 4: Top drivers of attrition

7. Risk Categories and HR KPIs

Out-of-fold probabilities were grouped into Low (< 0.3), Medium (0.3-0.6) and High (>= 0.6) risk. The table shows that the categories separate leavers well.

Risk category	Employees	Actual attrition rate
Low	691	0.042
Medium	450	0.136
High	329	0.447

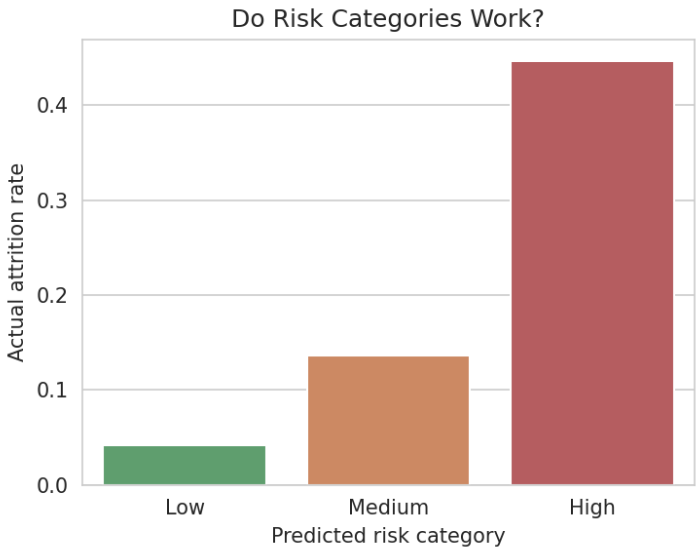


Figure 5: Actual attrition rate by predicted risk category

Company KPIs: attrition rate 16.1%, retention rate 83.9%, 329 employees currently in the High-risk category.

Department	Headcount	Leavers	Attrition rate	Retention rate	High-risk employees
Human Resources	63.000	12.000	0.191	0.809	16.000
Research & Development	961.000	133.000	0.138	0.862	181.000
Sales	446.000	92.000	0.206	0.794	132.000

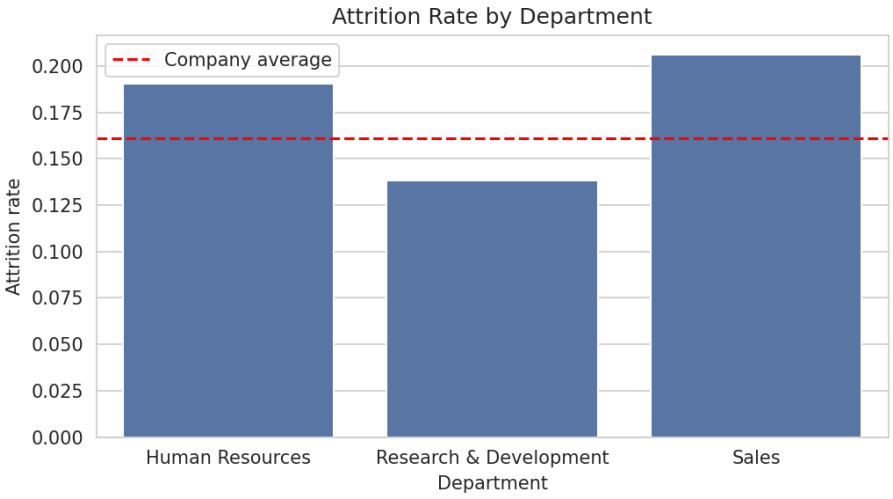


Figure 6: Attrition rate by department

The dataset has no time column, so department-wise trends are shown by tenure band (years at company) within each department.

8. Deployment on IBM Watson Machine Learning

- The model is a scikit-learn Pipeline (preprocessing + classifier), so raw employee records can be sent directly to the endpoint.
- Part 2 notebook: retrain -> evaluate -> quality gate (AUC >= 0.78) -> store model in Watson Machine Learning -> create or update the online deployment -> test scoring.
- Automation: the notebook is run as a Watson Studio Job on a schedule, with the API key and space ID as environment variables. Re-deployments update the same deployment so the endpoint URL does not change.
- Deployment evidence to add after you deploy: endpoint URL, deployment ID, a screenshot of the deployment space and a screenshot of a Job run.

Endpoint URL: <paste your scoring URL here after deployment>

9. Limitations and Future Enhancements

- Small, fictional dataset (1470 employees, 237 leavers): results will vary with the split and may not transfer to a real company.
- Feature importance shows association, not cause; predictions should support supportive HR conversations, never punitive decisions.
- Sensitive attributes (age, gender, marital status) need fairness checks before production use.
- Future work: monitoring for model drift, SHAP explanations in the dashboard, cost-based threshold selection, and integration with the HRIS.

Appendix A. REST API Documentation

Authentication: exchange the IBM Cloud API key for an IAM bearer token.

POST https://iam.cloud.ibm.com/identity/token
Content-Type: application/x-www-form-urlencoded
Body: apikey=<API_KEY>&grant_type=urn:ibm:params:oauth:grant-type:apikey

Scoring request:

POST
https://<region>.ml.cloud.ibm.com/ml/v4/deployments/<deployment_id>/predictions?version=2021-05-01
Headers: Authorization: Bearer <token>, Content-Type: application/json

Request body (one employee; send several rows in 'values' to score a batch):

```
{
 "input_data": [
  {
   "fields": [
    "Age",
    "BusinessTravel",
    "DailyRate",
    "Department",
    "DistanceFromHome",
    "Education",
    "EducationField",
    "EnvironmentSatisfaction",
    "Gender",
    "HourlyRate",
    "JobInvolvement",
    "JobLevel",
    "JobRole",
    "JobSatisfaction",
    "MaritalStatus",
    "MonthlyIncome",
    "MonthlyRate",
    "NumCompaniesWorked",
    "OverTime",
    "PercentSalaryHike",
    "PerformanceRating",
    "RelationshipSatisfaction",
    "StockOptionLevel",
    "TotalWorkingYears",
    "TrainingTimesLastYear",
    "WorkLifeBalance",
    "YearsAtCompany",
    "YearsInCurrentRole",
    "YearsSinceLastPromotion",
    "YearsWithCurrManager"
   ],
   "values": [
    [
     "41",
     "Travel_Rarely",
     "1102",
     "Sales",
     "1",
     "2",
```

```
    "Life Sciences",
    "2",
    "Female",
    "94",
    "3",
    "2",
    "Sales Executive",
    "4",
    "Single",
    "5993",
    "19479",
    "8",
    "Yes",
    "11",
    "3",
    "1",
    "0",
    "8",
    "0",
    "1",
    "6",
    "4",
    "0",
    "5"
   ]
  ]
 }
]
}
```


Typical response: a 'predictions' list with fields ['prediction', 'probability']; prediction 1 = likely to leave, 0 = likely to stay, and probability = [P(stay), P(leave)]. Map P(leave) to risk categories with the thresholds above.

Fields: Age, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, MonthlyRate, NumCompaniesWorked, OverTime, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager.